Final Project Report

Site Selection of Tesla's Next Gigafactory

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Instructor:	**Prof. Jeff S Busch**
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PURPOSE:

The purpose of this project is to assemble a cross functional team (see appendix a for organizational structure) to determine and recommend an optimal location for Tesla's next Gigafactory by May 16, 2025. The goal is to find a site that maximizes production efficiency and cost-effectiveness, building on Tesla's existing manufacturing expertise and integrating the latest developments in logistics, transportation, and workforce training. This effort is designed to enhance production capabilities more swiftly than previous projects, contributing to Tesla's leadership in electric vehicle manufacturing and supporting the broader shift towards sustainable transportation.

OBJECTIVES:

The primary objective of this project is to meticulously analyze and pinpoint the most suitable location for Tesla’s forthcoming Gigafactory, with the project completion targeted by May 16, 2025, and an allocated budget of $143,400. This new facility is envisioned to significantly amplify Tesla's manufacturing output of electric vehicle batteries, energy storage systems, and solar products, thereby enabling the company to decrease the production costs and expand its market reach effectively.

In order to select the most strategic location, the report will delve into a comprehensive evaluation of various critical factors including the robustness of the local supply chain, the efficiency of logistic operations, adherence to the regulatory and permitting frameworks, the dynamics of the local labor market, the availability and reliability of energy resources, and the level of community support and potential opposition. Each of these elements play a pivotal role in determining the feasibility and success of the new Gigafactory.

SCHEDULE:

The overall schedule consists of 8 key deliverables, over the course of 12 months (see appendix B for work breakdown structure and appendix J for the baseline schedule).

The project management phase for the Tesla Gigafactory kicks off with defining the scope and objectives in May and June, paralleled by stakeholder identification to ensure all relevant parties are engaged from the beginning. This early phase also includes establishing the project timeline and budget, ensuring foundational elements are set for subsequent activities. By July, the focus shifts to developing a communication cadence, crucial for maintaining project alignment and transparency.

As we transition into facility design in July and August, the project team will create the site layout and confirm the square footage, fundamental steps to ensure the facility meets Tesla's

operational needs. The drafting of the blueprint extends into September, with a pivotal step in August for obtaining LEED approval, underscoring Tesla's commitment to sustainability.

Site selection criteria are meticulously handled from August to December. Initially, the project evaluated the proximity to raw materials in August, followed by assessing renewable energy sources in September, ensuring the Gigafactory's operational efficiency and environmental alignment. The documentation of skilled workforce criteria in October and infrastructure and logistics in October and November sets the groundwork for regulatory and environmental considerations, culminating in November and December.

The actual selection of the location is refined through comprehensive reports and analyses from January to February. Formulating the site selection report and conducting a cost-benefit analysis ensure that the chosen location aligns with Tesla's strategic and financial targets, while a detailed risk assessment in February safeguards against potential future challenges.

Stakeholder engagement spans from July to October, starting with town hall meetings and extending to forming partnerships with local educational institutions in August and securing partnerships with the USDA and EPA by September. This phase is crucial for garnering local support and ensuring regulatory compliance.

Permit approvals are concentrated in the March and April timeline, covering federal, state, and county levels to address all regulatory requirements and ensure the project proceeds without legal hindrances.

The financial analysis phase in April and May focuses on detailing construction costs, operating expenses, and tax benefits. This crucial step not only ensures financial feasibility but also prepares the project for any economic contingencies.

Finally, the project closure activities in May involve documenting lessons learned and finalizing all project deliverables and reports. This last phase is vital for reviewing project outcomes and informing future projects, ensuring continuous improvement in Tesla's expensive efforts.

RESOURCES:

Over the course of the 12 month project, a total of 857 labor hours (see appendix L for resource loading table) split between the 8 project team members will be required in order to complete. Understanding that each resource is shared amongst the enterprise, is the primary driver of the project duration; ensuring not to put undue burden / constraint on the resources required for other strategic projects.

The project cost was estimated utilizing the bottoms up method to calculate both the labor and non-labor costs. The overall project cost is estimated at $143,400 and is split between three aspects (see appendix K for cost estimate):

1. **Labor ($42k):** Estimate took the labor rate of each resource and multiplied it by the hours required to complete each task (see appendix L for resource loading table).

2. **Non-Labor ($37.5k):** This is the non-recurring cost related to activities such as travel, permit funding and attendance fees for stakeholder engagement / networking events.
3. **Risk Mitigation / Buffer ($64k):** In order to mitigate the risks identified (see appendix H for risk matrix and appendix I for Issues, Impediments, Problems, Assumptions & Risks Log), the team has allocated funding to complete said activities. This also allows for a buffer to the overall time phased budget (see appendix M) in order to reduce the amount of disruptions and delays caused by additional funding requests and/or project financial adjustment.

PERSONNEL:

The project team will consist of 8 focal across the different critical functions that are also stakeholders of the new facility (see appendix D for stakeholder register and appendix G for the responsibility matrix). Brief summary of each stakeholder required:

Doug, Engineering: Accountable for creating project scope & objectives and responsible for drafting blueprints.

Sven, Operations Director: Accountable for establishing timeline & budget and responsible for site layout development.

Ryan, Finance: Accountable for financial analysis and developing the financial model.

Tonya, HR: Consulted on hiring staff and informed about workforce management.

Camille, Supply Chain: Accountable for securing raw materials and responsible for managing supply chain logistics.

Russel, Automotive GM: Consulted on site design and informed about engineering adjustments.

Mary, Compliance Manager: Accountable for obtaining all necessary permits and responsible for regulatory compliance.

Tyler, PM: Accountable for overall project management and responsible for stakeholder coordination.

RISK MANAGEMENT:

The Tesla Gigafactory project entails several identified risks that could impact the timeline, budget, and overall success of the project. 5 of which require mitigation plans which have been identified within the risk matrix (see appendix H). The risk management strategy is designed to proactively address these concerns through identification, assessment, and mitigation strategies. Here are the key risks along with the planned mitigation measures:

Failure to secure local suppliers and logistics: If the team fails to secure local suppliers during the project, then the flow of materials will be impacted, and a longer production lead time is the result.

Mitigation: Send a wider reaching request for proposal (RFP) for prospective suppliers

Failure to secure the proper permits in order to remain compliant and allow for construction: If the team fails to secure the proper permits for construction, then the company will be impacted by fines as a result.

Mitigation: Assign a compliance focal to prioritize all permit approvals.

Failure to secure skilled labor: If the team fails to identify skilled labor to staff the site, then the site will have poor quality output, resulting in increase to COGS and lead time

Mitigation: HR to secure partnerships with neighboring universities and trade schools to ensure a talent pipeline for the new facility

Financial mismanagement: If the budget is not managed properly, additional dollars may be required, resulting in additional costs for Tesla.

Mitigation: Ensure accurate cost estimation, secure sufficient funding, and manage cash flow. Add $50,000 to buffer for budget deviation.

Unexpected public pushback from local communities and environmental groups: If the local community does not welcome the new facility, then operations may be impacted, resulting in delays to production and delivery.

Mitigation: Extensive community engagement to address concerns and gather public support for the project

Each risk item and its mitigation plan will be reviewed as part of the recurring review cadence established within the monitor and control system.

CONTROL SYSTEM:

Baseline Execution Index (BEI): The primary control system will be a method in which to monitor the overall performance to schedule task execution. This will be accomplished by maintaining a baseline execution index (BEI), which measures the ratio of the total number of tasks that have been completed per the baselined schedule, compared to the actual.

Cadence: The BEI will be monitored on a weekly basis by the PM and recorded as one of the KPI's visible to the PMO and other functional leaders. PM will report to PMO management on a weekly basis to status performance to plan, and elevate risks / help needed. A monthly status meeting which includes all functional leaders and project sponsor will also be established

Threshold / KPI: In order to signal risk and/or flag the need for additional resources, the KPI will have a scoring criteria which indicates overall performance to plan. Scoring of deviation consists of the following:

Green	99% or higher BEI	On plan per the baseline schedule
Yellow	98% to 95% BEI	Off schedule with recovery plan in place
Red	94% or lower BEI	Off schedule with no recovery plan

All deviations are required to have a get to green plan established and shared in the recurring status meeting. Any scheduled re-baseline will be taken through the Change Control Board (CCB) steering committee which includes representatives from all functions involved in the project.

Financial Expenditure Report: To ensure performance to budget, the finance focal will maintain a weekly expenditure report to compare actuals for each task to the baseline time phased allocation of project budget. It will follow the same type of BEI methodology to track performance, with the difference being the allowed deviation of $10,000.

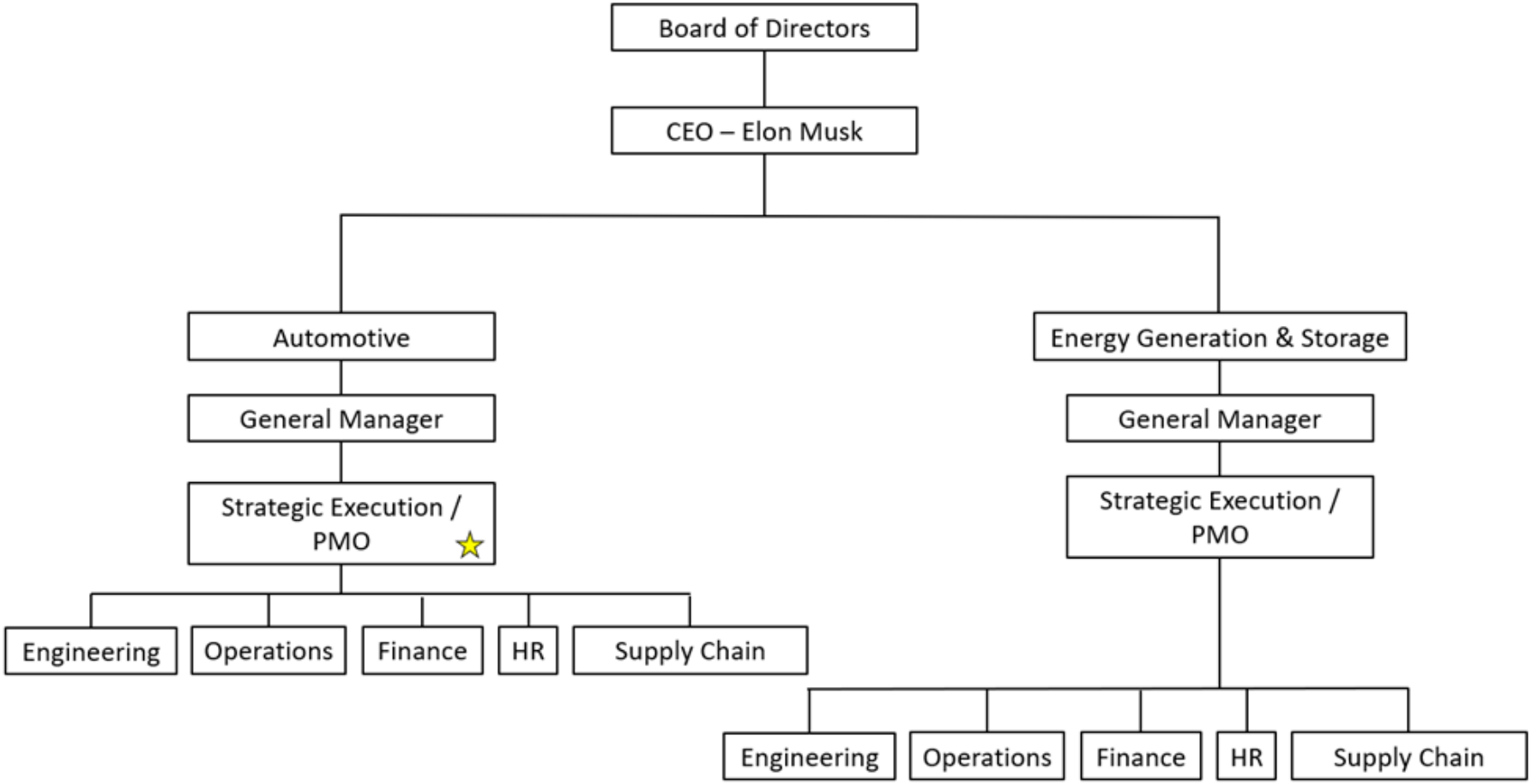
Cadence: This report will be reviewed weekly between the finance focal and PM. This KPI will also be included in the weekly and monthly status meetings happening with the PMO department, along with the other functional leaders.

TERMINATION PLAN:

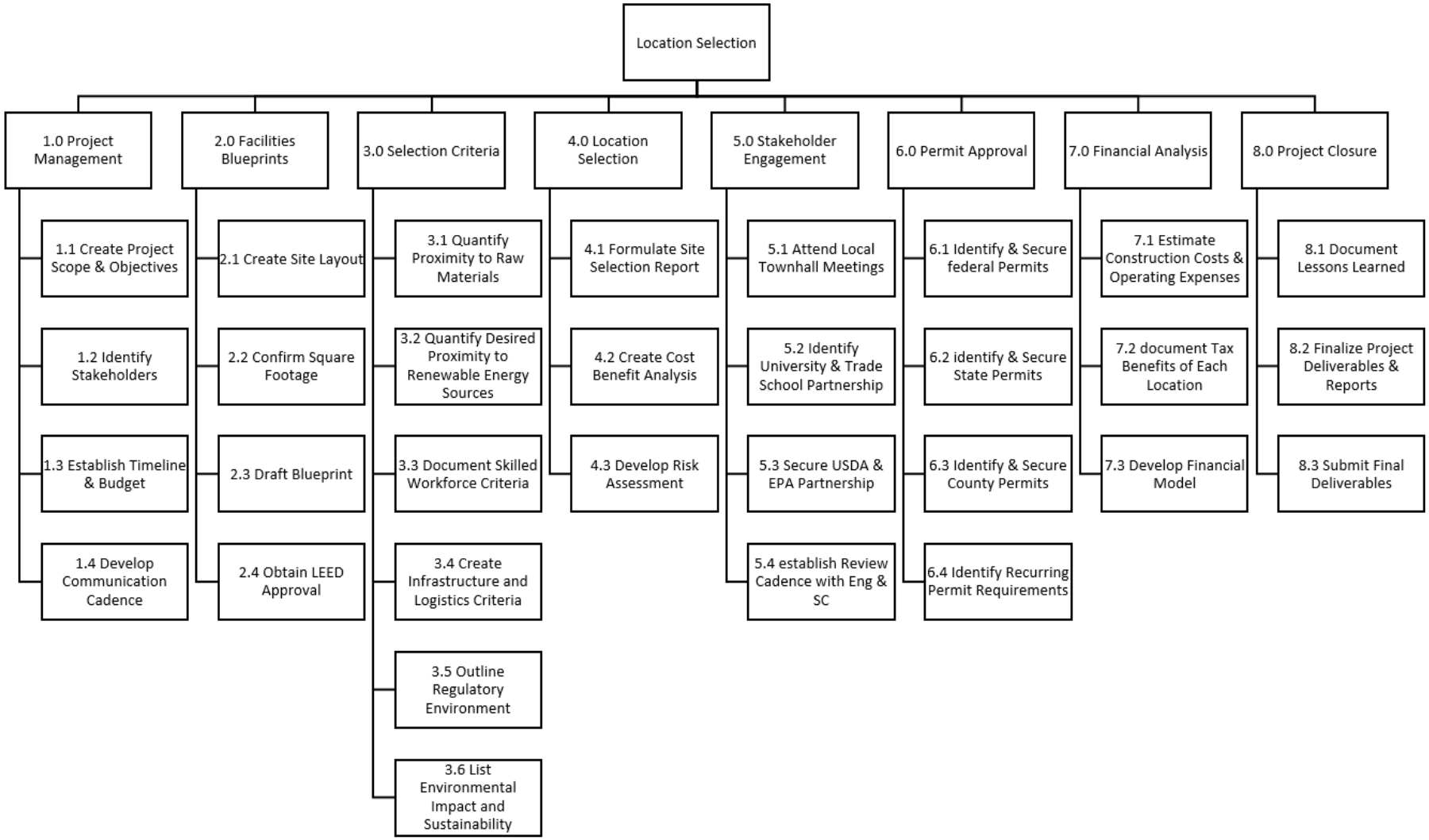
Once all closure criteria have been met, a final review will take place via the enterprise Change Control Board (CCB) for final project closure approval. Each task will be validated within this review (see appendix O for closure criteria) to ensure buyoff from each functional leader. After that point in time, procurement will proceed with the purchase of the proposed location.

Appendix

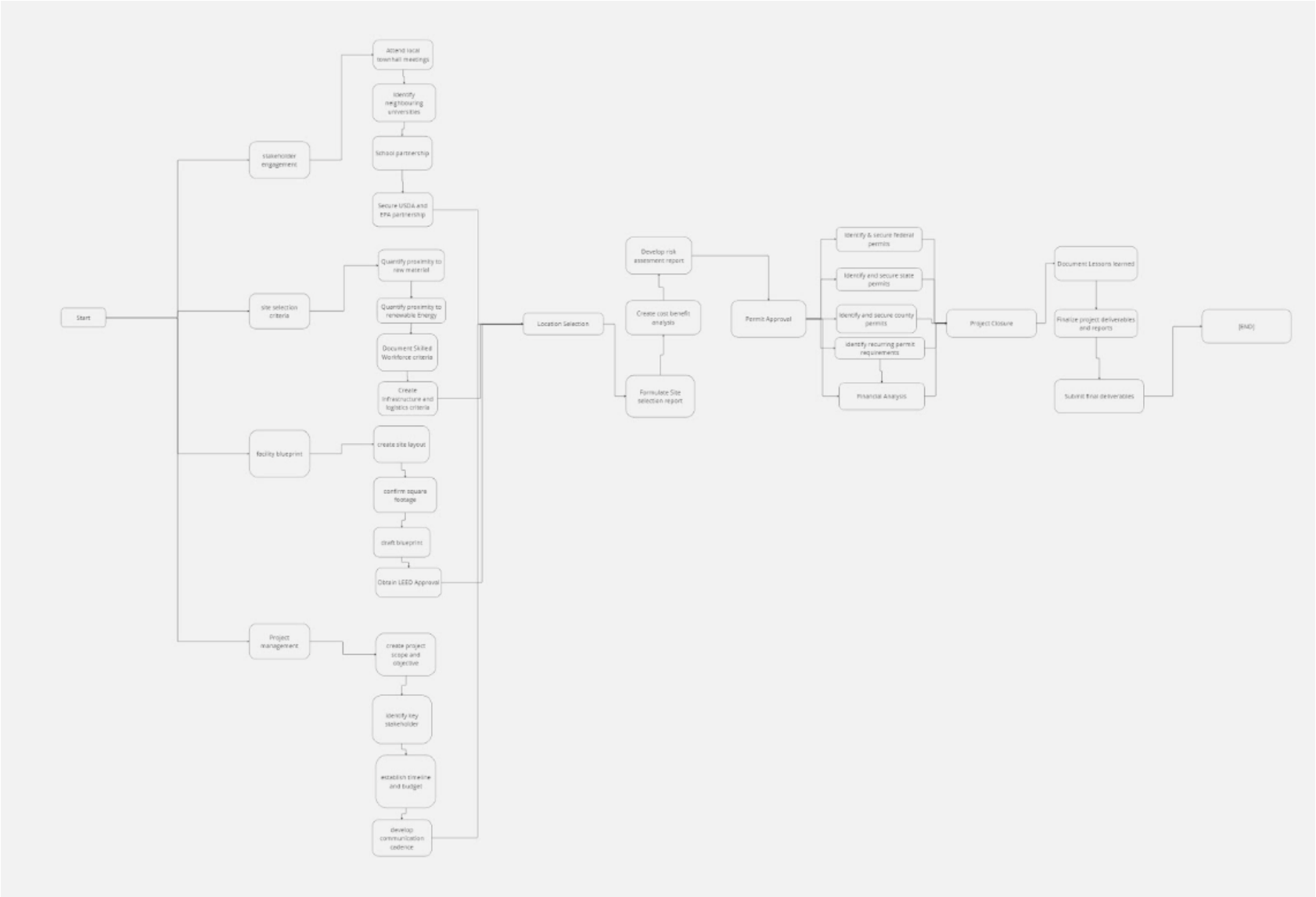
Appendix A: Organizational Breakdown



Appendix B: Work Breakdown Structure



Appendix C: Network Diagram

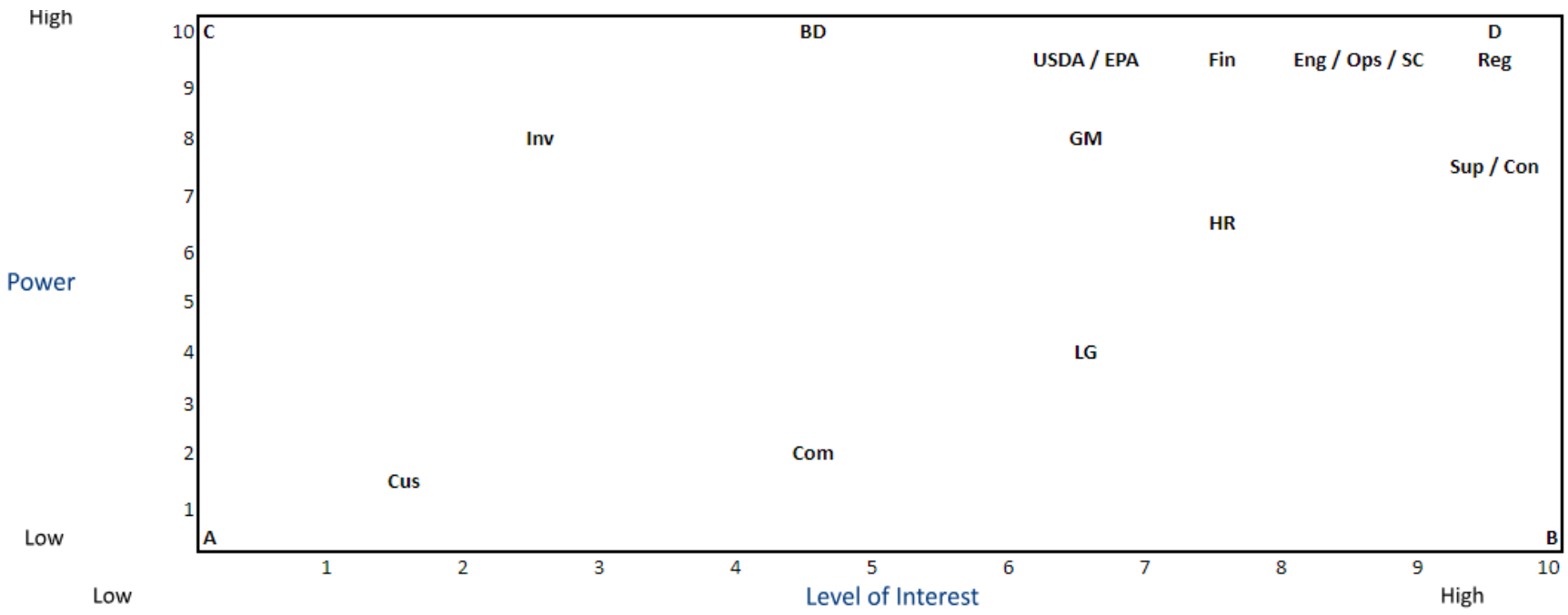


Appendix D: Stakeholder Register

Name	Org / Dept	Stakeholder Role	Impact / Influence	How to Engage?
Com	Communities	Resides in area of the new facility; impacted by all the indirect outcomes	Medium- Level of interest varies by the individual and proximity to the location of the facility. Their influence will be via local government. However, they contribute to the skilled labor required	Local town hall meetings for construction related issues College / Trade career fairs for engaging with skilled labor
Cus	Customers	End users who determine demand and products success	Low- Engagement with end product customers is outside scope of this internal company project	Advertisements; outside the scope of the project.
Inv	Investors / Shareholders	Provide the company with capital to fund projects / operations	Medium- Tesla currently has a high level of free cash flow in order to fund this project, however the continued engagement with investors is a key component to the future success of Tesla	CEO / Board of Directors engage with investors and shareholders. Out of scope for project team's level of engagement
Michael	US Department of Agriculture (USDA)	USDA Auditor: Responsible for the approval of development in rural areas and impacts to natural resources	High- Location selection is dependent upon the compliance with federal laws/ regulations. Approvals will be required for final location selection	Construction request submittal; potential survey or audit may be conducted by the department. Compliance focal must be assigned
Steve	Environmental Protection Agency (EPA)	EPA Auditor: Responsible for the approval of development based on impacts to environment	High- Location selection is dependent upon the compliance with federal laws/ regulations. Approvals will be required for final location selection	Construction request submittal; potential survey or audit may be conducted by the department. Compliance focal must be assigned
Richard	Local Government	Governor: Provides bids (grants) to have factory built in their region in order to produce additional jobs and revenue	Medium- Local government can provide grants and tax breaks as an incentive to bring Tesla's new facility to their region	Local town hall meetings and meetings with leadership to secure grant funding
Sup	Suppliers	Provide the material required in order to construct the facility	Medium- Critical aspect of the project will be managed by the supply chain team	Supply chain will have regular, recurring engagement throughout the project until all material has been obtained and is of proper quality

Hannah	3rd Party Construction	Construction Company Rep: Plan, coordinate, budget, and supervise construction project from start to finish, they monitor project and make decisions about construction activities	High- Project dependant on lead time for constructing the facility	Initial contract negotiations, then weekly project schedule development collaboration. Weekly progress to plan meetings during implementation
Doug	Engineering	Facilities Engineer: Responsible for designing the infrastructure of the facility	High- Design plans will be part of the project's critical path	Primary member of project team; weekly recurring meetings + ad hoc as needed
Sven	Operations	Operations Director: Responsible to mapping the layout of the building to ensure a lean / optimized value stream	High- Facility layout will be part of the project's critical path	Primary member of project team; weekly recurring meetings + ad hoc as needed
Ryan	Finance	Financial Analyst: Provides an estimate of project budget in order to seek approval. Track process and deviations	High- Key stakeholder in quantifying capital required and tracking adherence to budget	Primary member of project team; weekly recurring meetings + ad hoc as needed
Toyna	HR	HR Generalist: Responsible for hiring and setting up compensation of all new employees required for new facility	Medium- List of skills needed and hiring plan for new facility employees will be part of the project's critical path	Primary member of project team; weekly recurring meetings + ad hoc as needed
Camille	Supply Chain	Procurement Agent: Responsible for ordering all material required to complete the construction of the new facility	High- Key stakeholder in securing the material required for the construction of the facility	Primary member of project team; weekly recurring meetings + ad hoc as needed
Russel	Automotive General Manager	General Manager: Provides guidance and input on projects. Will report up and out to next level of leadership	High- Primary customer of the project. 2nd tier of approval after functions. Next higher layer is CEO & board of directors	Monthly performance to plan report outs. Member of the Change Control Board when seeking additional funding or other resources
BD	Board of Directors	Allocate funding and other resources in order to complete the project	High- Final tier of approval	CEO will engage with Board of Directors
Mary	Regulatory Affairs	Compliance Manager: Responsible for being the liaison between Tesla and government regulators to ensure that the selected location and structures are compliant	High- Significant impact to the overall execution of the project. Government permits and approvals are a predecessor to any development / construction implementation	Primary member of project team; weekly recurring meetings + ad hoc as needed

Appendix E: Stakeholder Grid



Appendix F: Project Charter

Deliverables	Measures	Exclusions
Project Management	• 100% buyoff from Change Control Board (CCB) • Post sprint project scope reviews	• Detailed Construction Plan
Facility Blueprints	• Percentage Utilization of Square Footage	• Bill of Materials list (BOM) • Electrical / Plumbing details
Site Selection Criteria	• Supply Chain Efficiency Index (SCEI) • Community Engagement Score (CES)	• Locations with Limited Access to Renewable Energy Sources. • Areas with Unfavorable Regulatory Environments • Procurement of material
Location Selection	• Geographic suitability assessment • Completion of stakeholder consultations within 3 months • Site survey to determine available square footage • Risk assessment KPI's	• Avoid areas lacking essential infrastructure (e.g., transportation, utilities) • Avoid areas with insufficient skilled labor or high labor costs
Stakeholder Engagement	• Execution of weekly stakeholder reviews • Team effectiveness assessment (TEA)	• Post location selection, external stakeholder engagement
Permit Approval	• Milestone Completion of local, state and federal permits	• Post Site Selection Permit Fees • Future Mitigation Plan for Regulatory Changes
Financial Analysis	• ROI • Budget utilization • Execution of weekly budget review	• Risk / sensitivity analysis
Project Closure	• Schedule Baseline Execution Index (BEI)	• Oversight of construction execution • Oversight of site procurement

Appendix G: Responsibility Matrix (R= Responsible; C= Contributor)

ID	Task	Doug Engineering	Sven Operations Director	Ryan Finance	Tonya HR	Camille Supply Chain	Russel Automotive GM	Mary Compliance Manager	Tyler PM
1.0	**Project Management**								
1.1	Create Project Scope & Objectives	C	C			C	C	C	R
1.2	Identify Key Stakeholders	C	C			C		C	R
1.3	Establish Timeline & Budget		C	R			C		C
1.4	Develop Communication Cadence	C	C	C		C	C	C	R
2.0	**Facility Blueprints (Design)**								
2.1	Create Site Layout	R	C			C		C	C
2.2	Confirm Square Footage	R					C	C	C
2.3	Draft Blueprint	R					C	C	C
2.4	Obtain LEED Approval	C						R	C
3.0	**Site Selection Criteria**								
3.1	Quantify Desired Proximity to Raw Materials					R	C		C
3.2	Quantify Desired Proximity to Renewable Energy Sources	R					C	C	C
3.3	Document Skilled Workforce Criteria	C	C	C	R	C			
3.4	Create Infrastructure and Logistics Criteria	R	C	C		C			
3.5	Outline Regulatory Environment							R	C
3.6	List Environmental Impact and Sustainability	C	C			C		R	C
4.0	**Location Selection**								
4.1	Formulate Site Selection Report	C	C	C		C		C	R
4.2	Create Cost-Benefit Analysis	C	C	R		C			C
4.3	Develop Risk Assessment report	C	C	C		C		C	R
5.0	**Stakeholder Engagement**								
5.1	Attend Local Townhall Meetings		C				C	R	C
5.2	Identify Neighboring University & Trade School Partnersh	C	C	C	R	C	C		C
5.3	Secure USDA & EPA Partnership						C	R	C
5.4	Establish Review Cadence with Engineering and Supply C	C				C			R
6.0	**Permit Approval**								
6.1	Identify and Secure Federal Permits						C	R	C
6.2	Identify and Secure State Permints						C	R	C
6.3	Identify and Secure County Permits						C	R	C
6.4	Identify Recurring Permit Requirements						C	R	C
7.0	**Financial Analysis**								
7.1	Estimate construction costs & operating expenses	C	C	R		C	C	C	C
7.2	Document Tax Benefits for Each Location		C	R	C		C	C	C
7.3	Develop Financial Model	C	C	R	C	C	C		C
8.0	**Project Closure**								
8.1	Document lessons learned	C	C	C	C	C	C	C	R
8.2	Finalize project deliverables and reports	C	C	C	C	C	C	C	R
8.3	Submit Final Deliverables								R

Appendix H: Risk Matrix

Risk Description	Probability	Impact	Prob x Impact	Response	Action	Cost
Failure to secure local suppliers and logistics.	3	5	15	Mitigate	Send a wider reaching request for proposal (RFP) for prospective suppliers	$1,000
Failure to secure the proper permits in order to remain compliant and allow for construction	4	4	16	Mitigate	Assign a compliance focal to prioritize all permit approvals	$1,000
Failure to secure skilled labor	3	3	9	Transfer	HR to secure partnerships with neighboring universities and trade schools to ensure a talent pipeline for the new facility	$10,000
Inefficient access to energy infrastructure	1	4	4	Accept	Energy proximity criteria built into plan	
Changes in market demand for electric vehicles	2	5	10	Accept	Conducting thorough market analysis and maintaining flexibility in production capabilities are important.	
Financial mismanagement	5	5	25	Avoid	Ensure accurate cost estimation, secure sufficient funding, and manage cash flow. Add $50,000 to buffer for budget deviation.	$50,000
Unexpected public pushback from local communities and environmental groups.	3	3	9	Mitigate	Extensive community engagement to address concerns and gather public support for the project	$2,000
Project not approved by Tesla's internal investment board	3	5	15	Accept	Create a comprehensive analysis for the board to consider all variables	

Appendix I: Issues, Impediments, Problems, Assumptions & Risks Log

DESCRIPTION	DATE	ASSIGNED TO	STATUS
Supply Chain and Logistics: Building a Gigafactory requires a robust supply chain to ensure a steady flow of raw materials, components, and equipment. Tesla would need to establish relationships with local suppliers and logistics partners to ensure efficient operations and minimize transportation costs.	4/21/2024	Camille- Supply Chain	Open
Regulatory and Permitting Processes: Establishing a large-scale manufacturing facility like a Gigafactory requires navigating complex regulatory and permitting processes. Tesla would need to comply with local laws, environmental regulations, and obtain necessary permits, which can be time-consuming and involve bureaucratic hurdles	4/21/2024	Mary- Regulatory Affairs	Open
Labor Laws and Workforce: Every country has unique labor market with specific labor laws and regulations. Tesla would need to navigate these laws, including regulations related to working hours, employee benefits, and labor unions. Adapting to the local workforce and ensuring a skilled labor pool for manufacturing operations could also pose challenges.	4/21/2024	Mary- Regulatory Affairs	Open
Energy Infrastructure: Gigafactories require a significant amount of energy to operate. Tesla would need to assess the availability and reliability of the local energy infrastructure in the country to ensure uninterrupted power supply for its manufacturing operations. This may involve negotiating energy contracts and potentially investing in renewable energy sources.	4/21/2024	Doug - Engineering	Open
Public Perception and Opposition: Large-scale industrial projects can sometimes face opposition from local communities or environmental groups concerned about the impact on the environment, land use, or other factors. Tesla would need to engage in effective stakeholder management, address concerns, and communicate the benefits of the Gigafactory to gain public support.	4/21/2024	Russel- General Manager	Open
Sustainability and Environmental Impact: Building and operating a Gigafactory can have significant environmental impacts, such as carbon emissions, water usage, and waste generation, requiring Tesla to implement sustainable practices and mitigate environmental harm.	4/21/2024	Steve- EPA	Open

Appendix J: Baseline Schedule

Task #	Task / Deliverable	Duration (m-days)	Start	Finish	Predecessor	Successor	Resource
1.0	**Project Management**		**6/3/2024**	**6/21/2024**			
1.1	Create Project Scope & Objectives	5 days	6/3/2024	6/7/2024		1.2, 1.3, 1.4	Tyler- PM
1.2	Identify Key Stakeholders	3 days	6/10/2024	6/13/2024	1.1	1.3, 5.4	Tyler- PM
1.3	Establish Timeline & Budget	5 days	6/14/2024	6/18/2024	1.1, 1.2	1.4	Ryan- Finance
1.4	Develop Communication Cadence	2 days	6/19/2024	6/21/2024	1.1, 1.2, 1.3	2	Tyler- PM
2.0	**Facility Blueprints (Design)**		**6/24/2024**	**8/2/2024**			
2.1	Create Site Layout	15 days	6/24/2024	7/9/2024	1	2.2, 2.3, 2.4	Doug- Engeering
2.2	Confirm Square Footage	2 days	7/10/2024	7/12/2024	2.1	2.3, 2.4	Doug- Engeering
2.3	Draft Blueprint	5 days	7/15/2024	7/19/2024	2.1, 2.2	2.4	Doug- Engeering
2.4	Obtain LEED Approval	10 days	7/22/2024	8/2/2024	2.3	3	Mary- CM
3.0	**Site Selection Criteria**		**8/5/2024**	**8/23/2024**			
3.1	Quantify Desired Proximity to Raw Materials	5 days	8/5/2024	8/9/2024	1.0, 2.0	4	Camille- SC
3.2	Quantify Desired Proximity to Renewable Energy Sources	5 days	8/5/2024	8/9/2024	1.0, 2.0	4	Doug- Engineering
3.3	Document Skilled Workforce Criteria	5 days	8/5/2024	8/9/2024	1.0, 2.0	4	Tonya- HR
3.4	Create Infrastructure and Logistics Criteria	10 days	8/5/2024	8/16/2024	1.0, 2.0	4	Doug- Engineering
3.5	Outline Regulatory Environment	10 days	8/5/2024	8/16/2024	1.0, 2.0	3.6, 4.0, 5.3	Mary- CM
3.6	List Environmental Impact and Sustainability	5 days	8/19/2024	8/23/2024	1.0, 2.0, 3.5	4	Mary- CM
4.0	**Location Selection**		**8/26/2024**	**10/11/2024**			
4.1	Formulate Site Selection Report	10 days	8/26/2024	9/6/2024	1.0, 2.0, 3.0	4.2	Tyler- PM
4.2	Create Cost-Benefit Analysis	15 days	9/9/2024	9/27/2024	4.1	4.3	Ryan- Finance
4.3	Develop Risk Assessment report	10 days	9/30/2024	10/11/2024	4.2	5	Tyler- PM
5.0	**Stakeholder Engagement**		**6/14/2024**	**11/6/2024**			
5.1	Attend Local Townhall Meetings	15 days	10/14/2024	11/2/2024	1.0, 2.0, 3.0, 4.0	6	Mary- CM
5.2	Identify Neighboring University & Trade School Partnership	3 days	11/4/2024	11/6/2024	5.1	5.1, 6.0	Tonya- HR
5.3	Secure USDA & EPA Partnership	10 days	8/19/2024	8/30/2024	3.5	6	Mary- CM
5.4	Establish Review Cadence with Engineering and Supply Chain	1 day	6/14/2024	6/17/2024	1.2	6	Tyler- PM
6.0	**Permit Approval**		**11/7/2024**	**3/7/2025**			
6.1	Identify and Secure Federal Permits	30 days	11/7/2024	12/19/2024	3.5, 5.2	6.2	Mary- CM
6.2	Identify and Secure State Permints	25 days	1/6/2025	1/31/2025	6.1	6.3, 6.4	Mary- CM
6.3	Identify and Secure County Permits	20 days	2/3/2025	2/28/2025	5.1, 6.2	6.4	Mary- CM
6.4	Identify Recurring Permit Requirements	5 days	3/3/2025	3/7/2025	6.1, 6.2, 6.3	7	Mary- CM
7.0	**Financial Analysis**		**3/10/2025**	**5/2/2025**			
7.1	Estimate construction costs & operating expenses	20 days	3/10/2025	4/4/2025	4.0, 5.0, 6.0	7.1	Ryan- Finance
7.2	Document Tax Benefits for Each Location	10 days	4/7/2025	4/18/2025	7.1	7.3	Ryan- Finance
7.3	Develop Financial Model	10 days	4/21/2025	5/2/2025	7.2	8.1	Ryan- Finance
8.0	**Project Closure**		**5/5/2025**	**5/16/2025**			
8.1	Document lessons learned	3 days	5/5/2025	5/7/2025	7	8.2, 8.3	Tyler- PM
8.2	Finalize project deliverables and reports	5 days	5/8/2025	5/15/2025	8.1	8.3	Tyler- PM
8.3	Submit FInal Deliverables	1 day	5/16/2025	5/16/2025	8.2		Tyler- PM

Appendix K: Cost Estimate

Assumptions:

1. Labor cost is calculated by applying the total hours for each task, multiplied by the specific labor rate of each contributor, found in the RMatrix table
2. As construction is out of scope for the project, no material costs are required. Therefore, total cost will be calculated by the sum of the overall labor
3. Final budget approval will be taken through all phases of the change control board (CCB) to ensure internal stakeholder / leadership approval

				BUDGET	ACTUAL	UNDER/OVER
				$ 143,391	$ -	$ -
	LABOR		NON-LABOR			
TASK	HRS	RATE		BUDGET	ACTUAL	UNDER/OVER
Project Management						
Create Project Scope & Objectives	44	$ 50.34		$ 2,215		$ (2,215)
Identify Key Stakeholders	31	$ 47.74		$ 1,480		$ (1,480)
Establish Timeline & Budget	28	$ 54.82		$ 1,535		$ (1,535)
Develop Communication Cadence	21	$ 49.90		$ 1,048		$ (1,048)
				$ 6,278	$ -	
Facility Blueprints (Design)						
Create Site Layout	36	$ 33.00		$ 1,188		$ (1,188)
Confirm Square Footage	6	$ 49.33		$ 296		$ (296)
Draft Blueprint	23	$ 48.35		$ 1,112		$ (1,112)
Obtain LEED Approval	18	$ 46.39	$ 2,000.00	$ 835		$ (835)
				$ 5,431	$ -	
Site Selection Criteria						
Quantify Desired Proximity to Raw Materials	20	$ 51.25	$ 10,000.00	$ 1,025		$ (1,025)
Quantify Desired Proximity to Renewable Energy Sources	25	$ 32.20	$ 10,000.00	$ 805		$ (805)
Document Skilled Workforce Criteria	45	$ 48.67	-	$ 2,190		$ (2,190)
Create Infrastructure and Logistics Criteria	25	$ 48.00	-	$ 1,200		$ (1,200)
Outline Regulatory Environment	15	$ 47.33	-	$ 710		$ (710)
List Environmental Impact and Sustainability	39	$ 47.18	-	$ 1,840		$ (1,840)
				$ 27,770	$ -	
Location Selection						
Formulate Site Selection Report	40	$ 48.13	-	$ 1,925		$ (1,925)
Create Cost-Benefit Analysis	50	$ 50.50	-	$ 2,525		$ (2,525)
Develop Risk Assessment report	55	$ 48.36	$ 5,000.00	$ 2,660		$ (2,660)
				$ 12,110	$ -	
Stakeholder Engagement						
Attend Local Townhall Meetings	33	$ 52.12	$ 1,500.00	$ 1,720		$ (1,720)
Identify Neighboring University & Trade School Partnership	49	$ 49.90	$ 1,500.00	$ 2,445		$ (2,445)
Secure USDA & EPA Partnership	17	$ 48.94		$ 832		$ (832)
Establish Review Cadence with Engineering and Supply Chain	29	$ 47.41		$ 1,375		$ (1,375)
				$ 9,372	$ -	
Permit Approval						
Identify and Secure Federal Permits	12	$ 52.08	$ 4,000.00	$ 625		$ (625)
Identify and Secure State Permints	12	$ 52.08	$ 2,000.00	$ 625		$ (625)
Identify and Secure County Permits	12	$ 52.08	$ 1,500.00	$ 625		$ (625)
Identify Recurring Permit Requirements	12	$ 52.08		$ 625		$ (625)
				$ 10,000	$ -	
Financial Analysis						
Estimate construction costs & operating expenses	51	$ 50.65		$ 2,583		$ (2,583)
Document Tax Benefits for Each Location	18	$ 52.89		$ 952		$ (952)
Develop Financial Model	17	$ 51.71		$ 879		$ (879)
				$ 4,414	$ -	
Project Closure						
Document lessons learned	38	$ 50.63		$ 1,924		$ (1,924)
Finalize project deliverables and reports	38	$ 51.10		$ 1,942		$ (1,942)
Submit Final Deliverables	3	$ 50.00		$ 150		$ (150)
				$ 4,016	$ -	
Risk Mitigation / Buffer			$ 64,000.00	$ 64,000		$ (64,000)
TOTAL				**$ 143,391**	**$ -**	

Appendix L: Resource Loading Table (By Resource)

Resource	Sum	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Engineering - Doug	**136**	**4**	**7**	**17**	**25**	**21**	**10**	**18**	**11**	**5**	**5**	**5**	**4**	**4**
Create Project Scope & Objectives	5	2	3											
Identify Key Stakeholders	4	2	2											
Develop Communication Cadence	15			15										
Create Site Layout	3				3									
Confirm Square Footage	3				3									
Draft Blueprint	15				7	8								
Obtain LEED Approval	5				2	3								
Quantify Desired Proximity to Renewable Energy Sources	10					10								
Document Skilled Workforce Criteria	5						5							
Create Infrastructure and Logistics Criteria	10						5	5						
List Environmental Impact and Sustainability	7							7						
Formulate Site Selection Report	2							1	1					
Create Cost-Benefit Analysis	10							5	5					
Develop Risk Assessment report	10								5	5				
Identify Neighboring University & Trade School Partnership	4		2	2										
Establish Review Cadence with Engineering and Supply Chain	10				10									
Estimate construction costs & operating expenses	10										5	5		
Develop Financial Model	2												2	
Document lessons learned	3												1	2
Finalize project deliverables and reports	3												1	2
Operations Director - Sven	**87**	**11**	**14**	**15**			**7**	**11**	**6**	**3**	**5**	**7**	**4**	**4**
Create Project Scope & Objectives	6	3	3											
Identify Key Stakeholders	6	3	3											
Establish Timeline & Budget	5	2	3											
Develop Communication Cadence	3			3										
Create Site Layout	10			10										
Document Skilled Workforce Criteria	5						5							
Create Infrastructure and Logistics Criteria	5						2	3						
List Environmental Impact and Sustainability	5							5						
Formulate Site Selection Report	2							1	1					
Create Cost-Benefit Analysis	5							2	3					
Develop Risk Assessment report	5								2	3				
Attend Local Townhall Meetings	6	3	3											
Identify Neighboring University & Trade School Partnership	4		2	2										
Estimate construction costs & operating expenses	10										5	5		
Document Tax Benefits for Each Location	2											2		
Develop Financial Model	2												2	
Document lessons learned	3												1	2
Finalize project deliverables and reports	3												1	2
Finance - Ryan	**91**	**6**	**7**	**4**			**12**	**14**	**14**	**3**	**7**	**13**	**7**	**4**
Establish Timeline & Budget	12	6	6											
Develop Communication Cadence	2			2										
Document Skilled Workforce Criteria	10						10							
Create Infrastructure and Logistics Criteria	5						2	3						
Formulate Site Selection Report	3							1	2					
Create Cost-Benefit Analysis	20							10	10					
Develop Risk Assessment report	5								2	3				
Identify Neighboring University & Trade School Partnership	3		1	2										
Estimate construction costs & operating expenses	15										7	8		
Document Tax Benefits for Each Location	5											5		
Develop Financial Model	5												5	
Document lessons learned	3												1	2
Finalize project deliverables and reports	3												1	2

Appendix L: Resource Loading Table (By Resource) Continued

Resource	Sum	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
HR - Tonya	**49**		**10**	**10**			**20**					**2**	**4**	**3**
Document Skilled Workforce Criteria	20						20							
Identify Neighboring University & Trade School Partnership	20		10	10										
Document Tax Benefits for Each Location	2											2		
Develop Financial Model	2												2	
Document lessons learned	3												1	2
Finalize project deliverables and reports	2												1	1
Supply Chain - Camille	83	6	7	10	5	10	7	14	8	3	2	3	4	4
Create Project Scope & Objectives	6	3	3											
Identify Key Stakeholders	6	3	3											
Develop Communication Cadence	3			3										
Create Site Layout	5			5										
Quantify Desired Proximity to Raw Materials	10					10								
Document Skilled Workforce Criteria	5						5							
Create Infrastructure and Logistics Criteria	5						2	3						
List Environmental Impact and Sustainability	7							7						
Formulate Site Selection Report	5							2	3					
Create Cost-Benefit Analysis	5							2	3					
Develop Risk Assessment report	5								2	3				
Identify Neighboring University & Trade School Partnership	3		1	2										
Establish Review Cadence with Engineering and Supply Chain	5				5									
Estimate construction costs & operating expenses	5										2	3		
Develop Financial Model	2												2	
Document lessons learned	3												1	2
Finalize project deliverables and reports	3												1	2
Automative GM - Russel	**67**	**8**	**11**	**5**	**2**	**12**				**6**	**7**	**5**	**6**	**5**
Create Project Scope & Objectives	6	3	3											
Establish Timeline & Budget	5	2	3											
Confirm Square Footage	1				1									
Draft Blueprint	3				1	2								
Quantify Desired Proximity to Raw Materials	5					5								
Quantify Desired Proximity to Renewable Energy Sources	5					5								
Attend Local Townhall Meetings	6	3	3											
Identify Neighboring University & Trade School Partnership	5		2	3										
Secure USDA & EPA Partnership	2			2										
Identify and Secure Federal Permits	3									3				
Identify and Secure State Permints	3									3				
Identify and Secure County Permits	3										3			
Identify Recurring Permit Requirements	3										3			
Estimate construction costs & operating expenses	3										1	2		
Document Tax Benefits for Each Location	3											3		
Develop Financial Model	2												2	
Document lessons learned	4												2	2
Finalize project deliverables and reports	5												2	3
Complaince Manager - Mary	**123**	**6**	**9**	**18**	**7**	**11**		**25**	**10**	**15**	**11**	**5**	**2**	**4**
Create Project Scope & Objectives	5	2	3											
Identify Key Stakeholders	5	2	3											
Develop Communication Cadence	3			3										
Create Site Layout	3			3										
Confirm Square Footage	1				1									
Draft Blueprint	2				1	1								
Obtain LEED Approval	10				5	5								
Quantify Desired Proximity to Renewable Energy Sources	5					5								
Outline Regulatory Environment	10							10						
List Environmental Impact and Sustainability	10							10						
Formulate Site Selection Report	10							5	5					
Develop Risk Assessment report	10								5	5				
Attend Local Townhall Meetings	5	2	3											
Secure USDA & EPA Partnership	12			12										
Identify and Secure Federal Permits	5									5				
Identify and Secure State Permints	5									5				
Identify and Secure County Permits	5										5			
Identify Recurring Permit Requirements	5										5			
Estimate construction costs & operating expenses	3										1	2		
Document Tax Benefits for Each Location	3											3		
Document lessons learned	3												1	2
Finalize project deliverables and reports	3												1	2

Appendix L: Resource Loading Table (By Resource) Continued

Resource	Sum	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Project Manager - Tyler	**221**	**24**	**29**	**16**	**17**	**14**		**27**	**23**	**18**	**10**	**6**	**18**	**19**
Create Project Scope & Objectives	16	8	8											
Identify Key Stakeholders	10	5	5											
Establish Timeline & Budget	6	3	3											
Develop Communication Cadence	5			5										
Create Site Layout	3			3										
Confirm Square Footage	1				1									
Draft Blueprint	3				1	2								
Obtain LEED Approval	3				1	2								
Quantify Desired Proximity to Raw Materials	5					5								
Quantify Desired Proximity to Renewable Energy Sources	5					5								
Outline Regulatory Environment	5							5						
List Environmental Impact and Sustainability	10							10						
Formulate Site Selection Report	15							7	8					
Create Cost-Benefit Analysis	10							5	5					
Develop Risk Assessment report	20								10	10				
Attend Local Townhall Meetings	16	8	8											
Identify Neighboring University & Trade School Partnership	10		5	5										
Secure USDA & EPA Partnership	3			3										
Establish Review Cadence with Engineering and Supply Chain	14				14									
Identify and Secure Federal Permits	4									4				
Identify and Secure State Permints	4									4				
Identify and Secure County Permits	4										4			
Identify Recurring Permit Requirements	4										4			
Estimate construction costs & operating expenses	5										2	3		
Document Tax Benefits for Each Location	3											3		
Develop Financial Model	2												2	
Document lessons learned	16												8	8
Finalize project deliverables and reports	16												8	8
Submit Final Deliverables	3													3

Appendix M: Time Phased Budget

Activities	Total Value	May	June	July	August	September	October	November	December	January	Feb	March	April	May	Buffer
Project Management	**$6,278**														
Create Project Scope & Objectives	$2,215	$1,108	$1,108												
Identify Key Stakeholders	$1,480	$740	$740												
Establish Timeline & Budget	$1,535	$768	$768												
Develop Communication Cadence	$1,048			$1,048											
Facility Blueprints (Design)	**$5,431**														
Create Site Layout	$1,188			$1,188											
Confirm Square Footage	$296				$296										
Draft Blueprint	$1,112				$556	$556									
Obtain LEED Approval	$2,835				$1,418	$1,418									
Site Selection Criteria	**$27,770**														
Quantify Desired Proximity to Raw Materials	$11,025					$11,025									
Quantify Desired Proximity to Renewable Energy Sources	$10,805					$10,805									
Document Skilled Workforce Criteria	$2,190						$2,190								
Create Infrastructure and Logistics Criteria	$1,200						$600	$600							
Outline Regulatory Environment	$710							$710							
List Environmental Impact and Sustainability	$1,840							$1,840							
Location Selection	**$12,110**														
Formulate Site Selection Report	$1,925							$963	$963						
Create Cost-Benefit Analysis	$2,525							$1,263	$1,263						
Develop Risk Assessment report	$7,660								$3,830	$3,830					
Stakeholder Engagement	**$9,372**														
Attend Local Townhall Meetings	$3,220	$1,610	$1,610												
Identify Neighboring University & Trade School Partnership	$3,945		$1,973	$1,973											
Secure USDA & EPA Partnership	$832			$832											
Establish Review Cadence with Engineering and Supply Chain	$1,375				$1,375										
Permit Approval	**$12,000**														
Identify and Secure Federal Permits	$4,625									$4,625					
Identify and Secure State Permints	$4,625									$4,625					
Identify and Secure County Permits	$2,125										$2,125				
Identify Recurring Permit Requirements	$625										$625				
Financial Analysis	**$4,414**														
Estimate construction costs & operating expenses	$2,583										$1,292	$1,292			
Document Tax Benefits for Each Location	$952											$952			
Develop Financial Model	$879												$879		
Project Closure	**$4,016**														
Document lessons learned	$1,924												$962	$962	
Finalize project deliverables and reports	$1,942												$971	$971	
Submit Final Deliverables	$150													$150	
Risk Mitigation / Buffer	**$64,000**														**$64,000**
Monthly Cummulative Value		$4,225	$6,198	$5,041	$3,645	$23,804	$2,790	$5,375	$6,055	$13,080	$4,042	$2,244	$2,812	$2,083	$145,391

Appendix N: Control Systems

Baseline Execution Index (BEI): The primary control system will be a method in which to monitor the overall performance to schedule task execution. This will be accomplished by maintaining a baseline execution index (BEI), which measures the ratio of the total number of tasks that have been completed per the baselined schedule, compared to the actual.

Cadence: The BEI will be monitored on a weekly basis by the PM and recorded as one of the KPI's visible to the PMO and other functional leaders. PM will report to PMO management on a weekly basis to status performance to plan, and elevate risks / help needed. A monthly status meeting which includes all functional leaders and project sponsor will also be established

Threshold / KPI: In order to signal risk and/or flag the need for additional resources, the KPI will have a scoring criteria which indicates overall performance to plan. Scoring of deviation consists of the following:

Green	99% or higher BEI	On plan per the baseline schedule
Yellow	98% to 95% BEI	Off schedule with recovery plan in place
Red	94% or lower BEI	Off schedule with no recovery plan

All deviations are required to have a get to green plan established and shared in the recurring status meeting. Any scheduled re-baseline will be taken through the Change Control Board (CCB) steering committee which includes representatives from all functions involved in the project.

Financial Expenditure Report: To ensure performance to budget, the finance focal will maintain a weekly expenditure report to compare actuals for each task to the baseline time phased allocation of project budget. It will follow the same type of BEI methodology to track performance, with the difference being the allowed deviation of $10,000.

Cadence: This report will be reviewed weekly between the finance focal and PM. This KPI will also be included in the weekly and monthly status meetings happening with the PMO department, along with the other functional leaders.

Appendix O: Project Closure Criteria

Definitions

Completion of Deliverables	Ensure all project objectives and deliverables are met.
Stakeholder Approval	Obtain formal acceptance from stakeholders.
Documentation	Finalize and archive all project documentation for future use.
Lessons Learned	Document lessons learned and best practices.
Financial Closure	Ensure all financial aspects of the project are settled.
Resource Release	Release project resources (team members, equipment, etc.).
Administrative Closure	Complete all administrative tasks related to project closure.
Post-Implementation Review	Conduct a review to evaluate project success and performance.
Final Reporting	Submit all final reports and deliverables.

List of Tasks for Project Closure

1. **Review Project Deliverables**
    a. Verify completion of all project deliverables.
    b. Ensure deliverables meet quality standards and requirements.
    c. Obtain formal acceptance from stakeholders.

2. **Documentation and Archiving**
    a. Collect and organize all project documents (plans, reports, emails, etc.). Make sure to include task status reports, actual completion dates of all contractual obligations.
    b. Archive project documentation in a central repository.
    c. Ensure access to archived documentation for future reference.
    d. Document final approved project scope.
    e. Document all environmental concerns and steps taken to offset any harm to the environment.

3. **Financial Closure**
    a. Review and finalize all project budgets and expenses.
    b. Ensure all invoices are paid and financial commitments are met.
    c. Reconcile project accounts and close them.

d. Prepare the final project financial report.

4. Resource Release

a. Release project team members and reassign them to other roles.
b. Return rented or borrowed equipment and resources.
c. Document the release and reallocation of resources.

5. Sprint Retrospective Meeting

a. Conduct a retrospective session with the project team and stakeholders.
b. Document lessons learned, including successes and areas for improvement.
c. Share lessons learned with the organization for future projects.

6. Administrative Closure

a. Complete all remaining administrative tasks (closing contracts, updating records, etc.).
b. Ensure all project files are properly labeled and stored.
c. Update the project management system to reflect project closure.

7. Post-Implementation Review

a. Schedule and conduct a post-implementation review meeting.
b. Evaluate project performance against initial objectives and KPIs.
c. Identify and document the project's impact and benefits realization.

8. Final Reporting

a. Prepare and submit the final project report.
b. Include an executive summary, project outcomes, financial summary, and lessons learned.
c. Present the final report to key stakeholders and the project sponsor